

Fundamental Function	Variables to Relate	Maxwell Relation
$U(S, V, N)$ $dU = TdS - pdV + \mu dN$	S,V S,N V,N	$(\frac{\partial T}{\partial V})_{S,N} = -(\frac{\partial p}{\partial S})_{V,N}$ $(\frac{\partial T}{\partial V})_{S,V} = (\frac{\partial \mu}{\partial S})_{V,N}$ $-(\frac{\partial p}{\partial N})_{S,V} = (\frac{\partial \mu}{\partial V})_{S,N}$
$F(T, V, N)$ $dF = -SdT - pdV + \mu dN$	T,V T,N V,N	$(\frac{\partial S}{\partial V})_{T,N} = (\frac{\partial p}{\partial T})_{V,N}$ $-(\frac{\partial S}{\partial N})_{T,V} = (\frac{\partial \mu}{\partial T})_{V,N}$ $-(\frac{\partial p}{\partial N})_{T,V} = (\frac{\partial \mu}{\partial V})_{T,N}$
$H(S, p, N)$ $dH = TdS + Vdp + \mu dN$	S,p S,N p,N	$(\frac{\partial T}{\partial p})_{S,N} = (\frac{\partial V}{\partial S})_{p,N}$ $(\frac{\partial T}{\partial N})_{S,p} = (\frac{\partial \mu}{\partial S})_{p,N}$ $(\frac{\partial V}{\partial N})_{S,p} = (\frac{\partial \mu}{\partial p})_{S,N}$
(S, V, μ) $TdS - pdV - Nd\mu$	S,V S, μ V, μ	$(\frac{\partial T}{\partial V})_{S,\mu} = -(\frac{\partial p}{\partial S})_{V,\mu}$ $(\frac{\partial T}{\partial \mu})_{S,V} = -(\frac{\partial N}{\partial S})_{V,\mu}$ $(\frac{\partial p}{\partial \mu})_{S,V} = (\frac{\partial N}{\partial V})_{S,\mu}$
$G(T, p, N)$ $dG = -SdT + Vdp + \mu dN$	T,p T,N p,N	$-(\frac{\partial S}{\partial p})_{T,N} = (\frac{\partial V}{\partial T})_{p,N}$ $-(\frac{\partial S}{\partial N})_{T,p} = (\frac{\partial \mu}{\partial T})_{p,N}$ $(\frac{\partial V}{\partial N})_{T,p} = (\frac{\partial \mu}{\partial p})_{T,N}$
(T, V, μ) $-SdT - pdV - Nd\mu$	T,V T, μ V, μ	$(\frac{\partial S}{\partial V})_{T,\mu} = (\frac{\partial p}{\partial T})_{V,\mu}$ $(\frac{\partial S}{\partial \mu})_{T,V} = (\frac{\partial N}{\partial T})_{V,\mu}$ $(\frac{\partial p}{\partial \mu})_{T,V} = (\frac{\partial N}{\partial V})_{T,\mu}$

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(S, p, μ) $TdS + Vdp - Nd\mu$	S, p S, μ p, μ	$(\frac{\partial T}{\partial V})_{S,\mu} = (\frac{\partial V}{\partial S})_{p,\mu}$ $(\frac{\partial T}{\partial \mu})_{S,p} = -(\frac{\partial N}{\partial S})_{p,\mu}$ $(\frac{\partial V}{\partial \mu})_{S,p} = -(\frac{\partial N}{\partial p})_{S,\mu}$
(T, p, μ) $-SdT + Vdp - Nd\mu$	S, p T, μ p, μ	$-(\frac{\partial S}{\partial p})_{T,\mu} = (\frac{\partial V}{\partial T})_{p,\mu}$ $(\frac{\partial S}{\partial \mu})_{T,p} = (\frac{\partial N}{\partial T})_{p,\mu}$ $(\frac{\partial V}{\partial \mu})_{T,p} = -(\frac{\partial N}{\partial p})_{T,\mu}$